Lecture 15: Attention

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1 Review of RNN++

- Prior knowledge: text can be long, early observations may be important, some information is meaningless, and a logic break should reset the state.

- GRU: the update gate interpolates between the candidate and the previous state, $H_t = (1 - Z_t) \odot \tilde{H}_t + Z_t \odot H_{t-1}$, with reset gates capturing short-term dependencies and update gates long-term ones.

- LSTM: the cell state $C_t = F_t \odot C_{t-1} + I_t \odot \tilde{C}_t$ adds a candidate instead of multiplying, and $H_t = O_t \odot \tanh(C_t)$ reads the memory.

- Deep RNN/LSTM/GRU: stacking layers increases capacity, with lower layers learning shorter-term dynamics and higher layers learning high-level trends.

- Bidirectional models see both directions but cannot do forward prediction and are very slow; they fit filling in missing words, token annotation, and sequence encoding.

- Seq2seq: an encoder compresses the input into a context $c = q(h_1, \ldots, h_T)$, and a decoder generates a variable-length output with cross-entropy per position.

2 Attention as neighbor pooling

- Machine translation with a single hidden state cannot guarantee a proper representation for each word; the state flows across the sentence and must aggregate everything.

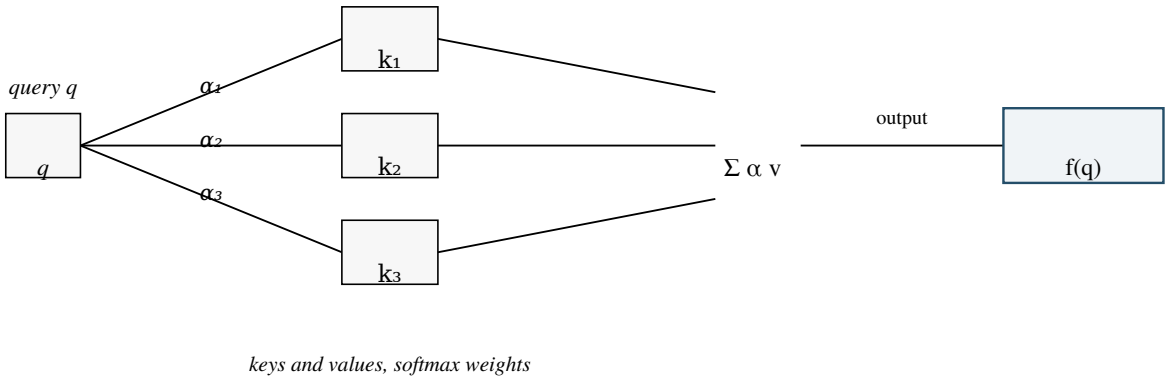


Figure 1. Attention pools the values through softmax weights: the query compares itself with every key, and the weighted values produce the output.

- Representation with neighbor information: an average $f(x) = (1/n)\Sigma y_i$, or a weighted average $f(x) = \Sigma \alpha(x, x_i) y_i$ with α related to similarity or distance. In the most general form, this is attention.

2.1 Attention for machine translation

- The hidden state of each word considers interactions with all other words directly; a query hidden state gathers them through $f(x) = \Sigma \alpha(x, x_i) y_i$.

- The output hidden state is used for the downstream task, while the predefined state of each token carries its context and neighbor information.

- Three questions: how to encode characters (keys)? how to define the values? how to define the attention weight function?

- Keys: one-hot encoding or a fully connected network over the one-hot vector. Values: a fully connected network over the one-hot vector.

- Attention weights (dot-product): $\alpha(q, k_i) = exp(a(q, k_i)) / \Sigma j_j$exp(a(q, k_j)) with $a(q, k) = q \top k / \sqrt{d}$.

- Loss function: cross-entropy for each word prediction; training uses the common optimizers with back-propagation.

2.2 Self-attention

- For each input $x_i$in the sentence, $y_i = \Sigma j_j \alpha(x_i, x_j) v_j$.

- Each output is a context-aware hidden state for downstream classification or other tasks, attending against the sentence itself — including the token itself.

- All tokens can be computed in parallel; encoding and embedding are the most important things for NLP.

2.3 Position encoding

- Attention pools without order information. Time series, sequences, and text all carry order, so positions must be injected explicitly.

- For $X \in \mathbb{R}$ n×d, a matrix $P \in \mathbb{R}$ n×dadds sine and cosine features with frequencies that decay along the dimension:

$$p_{i,2j} = \sin\left(\frac{i}{10000^{\frac{2j}{d}}}\right), \qquad p_{i,2j+1} = \cos\left(\frac{i}{10000^{\frac{2j}{d}}}\right).$$

- It looks like a weird function but people use it; it encodes absolute and relative positional information. The input dimension equals the raw encoding dimension plus the position encoding dimension.

2.4 Multi-head attention and the Transformer

- Input of an attention layer: L × (position embedding + one-hot); output: L × hidden-state dimension.
- To increase complexity: multi-head attention and stacking more attention layers.
- Multi-head attention projects queries, keys, and values into different representation subspaces, attends in each head, and concatenates the results.
- A Transformer block: multi-head attention learns the hidden state; fully connected layers control the dimension (for easy implementation); masked attention ignores padding positions; multiple blocks increase complexity.

3 Attention as a function approximator

- Fully-connected layers are a general function approximator: with enough nodes, layers, and non-linearity, any relation can be approximated (universal approximation theorem).
- A function is a mapping from a set X to a set Y assigning exactly one element of Y to each element of X. It can be represented by a mathematical formula or by a mapping table.
- Mathematical formula: memory efficient, but the form must be defined, and accuracy on the existing data depends on the form.
- Mapping table: precise for the existing data, but it must store the data, and predictions for unknown data are unclear — one can answer with neighbor information.
- Example: predict the gender of P5 (1.75 m, 70 kg) from a table of people; using neighbor information is the systematic way to make the prediction.

3.1 Nadaraya–Watson kernel regression

- Attention is related to nonvolitional cues based on saliency and volitional cues that depend on the task; pooling neighbors with weights is the core operation, $f(x) = \Sigma\, \alpha(x, x_i)\, y_i$.

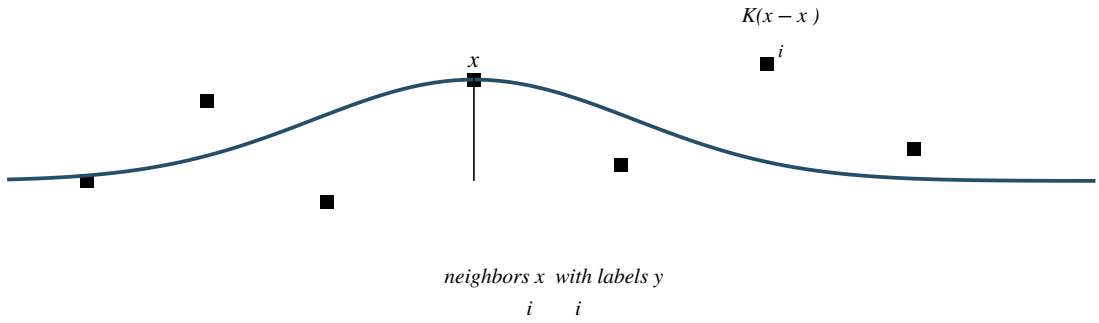


Figure 2. Nadaraya–Watson regression: a Gaussian kernel centered at the query x weights the nearby training points, and the normalized weights define the prediction.

- Nadaraya–Watson regression normalizes a kernel over all neighbors: $f(x) = \Sigma\, K(x - x_i)\, y_i / \Sigma\, j_j K(x - x_j)$.

$$f(x) = \sum_{i=1}^{n} \frac{K(x - x_i)}{\Sigma_j\, K(x - x_j)}\, y_i, \qquad K(u) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{u_2}{2})$$

- With the Gaussian kernel $K(u) = (1/\sqrt{2\pi}) \exp(-u^2/2)$, the prediction becomes a softmax of squared distances: $f(x) = \Sigma\, exp(-(x-x\, i_i)^2/2)\, y_i / \Sigma\, j_j exp(-(x-x\, j_j)^2/2)$.

$$f(x) = \sum_{i=1}^{n} \frac{\exp(-\frac{(x - x_i)^2}{2})}{\Sigma_j\, \exp(-\frac{(x - x_j)^2}{2})}\, y_i$$

- Attention with parameters: make the kernel width learnable, $f(x) = \Sigma\, exp(-(x-x\, i_i)^2\, w/2)\, y_i / \Sigma\, j_j$ $exp(-(x-x\, j_j)^2\, w/2)$. This is kernel smoothing whose band is fitted from data.

4 The attention pooling formula

- The commonly used attention computes an output as a weighted combination of values, with weights from a softmax over a score function:

$$f(q, k_1, v_1, \ldots, k_m, v_m) = \sum_{i=1}^{m} \alpha(q, k_i)\, v_i$$

$$\alpha(q, k_i) = \text{softmax}(a(q, k_i)) = \frac{\exp(a(q, k_i))}{\sum_j \exp(a(q, k_i))}$$

• Different score functions a lead to different attention behaviors.

4.1 Additive attention

• The score combines query and key through a small network, $a(q, k) = w\, v^T \tanh(\mathbf{W}_q q + \mathbf{W}_k k)$. It is usually used when queries and keys are vectors of different lengths.

$$a(q, k) = \mathbf{w}\, v^T \tanh(\mathbf{W}_{q_s} q + \mathbf{W}_{k_s} k)$$

4.2 Scaled dot-product attention

• The score is the dot product scaled by $\sqrt{d}$, $a(q, k) = q^T k / \sqrt{d}$. It is usually used when queries and keys are vectors of the same length.

$$a(q, k) = \frac{q^T k}{\sqrt{d}}$$

• How to improve it: multi-head attention projects queries, keys, and values into different representation subspaces and pools in each head.

4.3 Self-attention with position encoding

• Applied to a sequence, the embedding of each token is $y_i = f(x_i, x_1, x_1, \ldots, x_n, x_n)$.

• There is no order information in the pooling, so position encoding is added with the sine/cosine scheme of section 2.3.

5 Transformer

• Putting everything together: token embeddings plus position encoding, stacked blocks of multi-head self-attention and fully connected layers, with masked attention that ignores padding positions.

- After learning the hidden states (embeddings), the Transformer supports machine translation, token-level prediction, sentence-level prediction, text generation, and more.

- With labelled data it is supervised learning with cross-entropy loss.

- If we only have a corpus without labels, what should the loss be? This motivates self-supervised objectives such as masked language modeling and next-token prediction, the topic of the next lecture.